Q1.

(a) Magnesium chloride is formed when magnesium reacts with an acid.

(i) Complete the word equation for the reaction between magnesium and this acid.

magnesium + ...→ magnesium chloride + ...

2 marks

(ii) Suggest why magnesium chloride can be made by mixing magnesium with this acid but copper chloride **cannot** be made by mixing copper with this acid.

.....

.....

1 mark

(b) Copper sulphate is made by adding copper oxide to a different acid.
Give the name of the acid which is used.

.....

1 mark

(c) In the table below, write the name of the compound represented by each formula.

formula	**name**
$CuSO_4$	
$MgCl_2$	

2 marks

Maximum 6 marks

Q1.

(a) (i) *magnesium* + hydrochloric acid →

1 (L7)

→ magnesium chloride + hydrogen

*do **not** accept 'hydrogen chloride'*

*do **not** accept formulae*

1 (L7)

(ii) magnesium is more reactive than hydrogen **and** copper is less reactive than hydrogen

accept 'magnesium is more reactive than copper'

accept 'copper is less reactive than magnesium'

accept 'magnesium is higher than copper in the reactivity series'

accept 'copper is lower in the reactivity series'

1 (L7)

(b) sulphuric

1 (L7)

(c)

formula	name
$CuSO_4$	copper sulphate
$MgCl_2$	magnesium chloride

2 (L7)

[6]